Analysing Two-Way Tables

NCEA Level 2 Statistics — Bivariate Categorical Data

A **two-way table** sorts data by two categories at once. Whenever the **groups being compared are different sizes**, comparing raw counts can be misleading — you need to convert counts to **percentages** first.

Row percentage $= \dfrac{\text{cell value}}{\text{row total}} \times 100$. This tells you, *within one group*, what share falls into each category — which is exactly what you need to fairly compare groups of different sizes.

Worked example: 40 students at School A and 200 students at School B were asked if they play a sport. 28 students at School A said yes, and 120 students at School B said yes. More School B students said yes (120 vs 28) — but School B also asked far more students, so this comparison is misleading. Using row percentages instead:

$$\text{School A: } \frac{28}{40} \times 100 = 70\%, \qquad \text{School B: } \frac{120}{200} \times 100 = 60\%.$$

A *higher percentage* of School A students play a sport, even though fewer of them said yes in total.

1 Calculating Percentages

170 Year 13 students were asked how they studied for an exam, and whether they passed.

	Passed	**Failed**	**Total**
Group study	56	14	
Solo study	42	18	
Did not study	10	30	
Total			

1. Copy and complete the table, filling in all the totals.
2. What percentage of students who studied in a group passed?

   _____

3. What percentage of students who studied alone passed?

   _____

4. What percentage of students who did not study passed?

   _____

5. What percentage of *all* 170 students passed?

   _____

2 Comparing Distributions

Use your percentages from Section 1 to answer these.

1. Write a sentence comparing the pass rate of students who studied in a group with students who did not study, using percentages.

   _____

2. Tyler says, "Group study must be better than solo study, because more students who did group study passed (56) than students who did solo study (18 failed vs 14 failed)." Explain what is wrong with the way Tyler is comparing these two groups.

3. Using percentages, write a correct comparison between group study and solo study.

3 Describing a Trend

400 people, in four age groups, were asked which type of transport they use most often to get to work.

	Public transport	**Car**	**Other**	**Total**
15–24	45	30	25	
25–44	30	60	10	
45–64	20	70	10	
65+	15	55	30	

1. Copy and complete the table, filling in the total for each age group.

2. Calculate the percentage of each age group that uses public transport most often.

3. Describe the trend in public transport use as age increases. Refer to your percentages.

4. The percentage using "Other" transport is much higher for the 65+ age group than for the 45–64 age group. Suggest a reason this might be the case.

4 Making a Justified Conclusion

Two Year 12 classes were asked whether they like rugby.

	Like	**Dislike**	**Total**
Class A	14	8	
Class B	11	13	

1. Calculate the percentage of each class that likes rugby.

2. Class A's percentage is noticeably higher than Class B's. Explain why this difference might *not* show a real difference between the two classes, given how many students are in each sample.

3. What could you do to become more confident that there is a genuine difference between the two classes, rather than the difference being due to chance?

4. In Section 3, you found that public transport use decreases as age increases. This survey only sampled people from one city. Explain why you should be cautious about concluding this trend is true for *everyone*, everywhere.